Workshop: Concentrations

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Conversion between concentrations

1. Convert the following concentrations into mg/ml
   a. 100 mg in 5 ml
   b. 1000 mg in 20 ml
   c. 1 g in 100 ml
   d. 1 g in 200 ml
   e. 2 g in 500 ml
   f. 10 g in 1 l

   Which of the concentrations in question 1 is the weakest and which is the strongest?

2. Convert the following concentrations into %w/v. %w/v means g/100 ml.

$$\%\text{w/v} = \frac{\text{mass (g)}}{\text{volume (ml)}} \times 100$$

   a. 1 g in 100 ml
   b. 5 g in 100 ml
   c. 5 g in 1000 ml
   d. 5 g in 50 ml
   e. 500 mg in 10 ml
   f. 150 mg/ml

3. Convert each of the following concentrations into the given unit.
   a. 250 mg/10 ml into mg/ml
   b. 250 mg/10 ml into %w/v
   c. 40% w/v into mg/ml
   d. 2% w/v into mg/ml

4. Convert each of the following concentrations into %v/v. %v/v can be thought of as ml/100 ml (or l/100 l).
   a. 50 ml in 200 ml
   b. 2 ml in 50 ml
   c. 500 ml in 2 l
   d. 0.5 ml in 500 ml

5. Order the following concentrations from weakest to strongest.

   A: 1% w/v

   B: 300 mcg/ml

   C: 1 mg/5 ml

   D: 1 mg/ml

Calculating dosage

6. A patient requires a drug dose of 200 mg. The drug comes in liquid form with a strength of 5 mg/ml. How many ml does the patient need?

7. A patient requires a drug dose of 500 mg. The drug comes in liquid form with a strength of 2 % w/v. How many ml does the patient need?

8. A patient requires a drug dose of 2 g. The drug comes in liquid form with a strength of 800 mg/10 ml. How many ml does the patient need?

9. A patient is given a drug that comes in a strength of 0.4% w/v. The patient needs 20 mg of the drug, twice daily for 7 days. What is the total number of ml required for the treatment?

10. A particular drug comes in a strength of 10 % v/v. The patient needs an active dose of 5 ml per day for 5 days, followed by 10 days of 2.5 ml per day. What is the total ml needed to complete the treatment?